

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 hour

QUESTIONS: 2

Total Marks:40

Annexure A

Integrated Experiments

Code of Conduct:

I **Vempati Murale Karrtik (192312567)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Vempati Murale Karrtik

Signature of the student:

1. Aim and Problem Understanding

The aim of this integrated experiment is to develop an end-to-end intelligent maintenance pipeline for a manufacturing plant. Sensor measurements such as vibration, temperature, operating hours and machine load are first used to predict whether equipment is likely to fail. The resulting failure-risk state is then passed to a Q-Learning agent that recommends an appropriate maintenance action: Inspect, Lubricate, Replace Part, or Continue Operation.

2. Objectives

Implement a Decision Tree classifier and an MLP Neural Network using the same equipment dataset.

Evaluate both classifiers using Accuracy, Precision, Recall and F1-Score.

Visualize the first three levels of the Decision Tree and the MLP layer/neuron architecture.

Select a suitable model for real-time failure alerts using both numerical performance and deployment considerations.

Use classifier failure risk as the state input to a Q-Learning maintenance agent.

Train the agent over simulated machine episodes, obtain the final Q-table and extract the learned policy.

Demonstrate an end-to-end pipeline from sensor data to risk prediction and maintenance recommendation.

3. Dataset Description

A reproducible synthetic equipment dataset containing 500 machine records was generated for demonstration because no plant dataset was supplied. The target variable Failure is binary: 0 represents normal operation and 1 represents a failure condition. The same training and test partitions are used for both supervised learning models.

Feature	Type	Meaning
Vibration	Continuous	Machine vibration level
Temperature	Continuous	Operating temperature in °C
Operating Hours	Continuous	Accumulated machine operating hours
Load	Continuous	Machine load percentage
Failure	Binary	0 = No Failure, 1 = Failure

4. Experimental Methodology

Split the dataset into 75% training and 25% testing using stratification.

Train a Decision Tree with controlled depth to improve interpretability.

Standardize inputs and train an MLP with 16 and 8 neurons in two hidden layers.

Generate failure probabilities and convert them into Low, Medium and High risk states.

Use the risk state as the Q-Learning state and select one of four maintenance actions.

5. Question 1 – Decision Tree Classifier

The Decision Tree classifier recursively partitions the sensor feature space into regions associated with failure and non-failure classes. A maximum depth of five was used for the classifier, while only the first three levels are shown in the report for readability. Class balancing was enabled so that the classifier does not ignore the failure class.

The model was trained using Vibration, Temperature, Operating Hours and Load. After training, predictions were generated for the held-out test set and evaluated using four standard classification metrics.

6. MLP Neural Network Classifier

The Neural Network is a Multi-Layer Perceptron consisting of four input neurons, two hidden layers containing 16 and 8 neurons respectively, and one output neuron. ReLU activation is used in the hidden layers and the output represents the failure class probability. StandardScaler is applied before the MLP so that the input variables operate on comparable numerical scales.

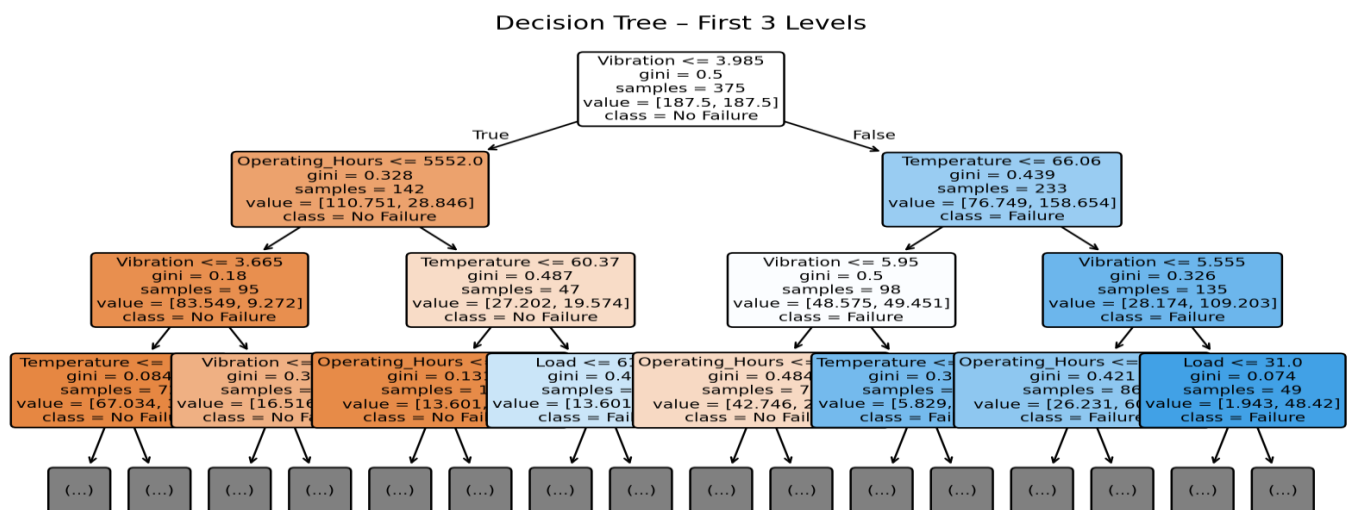


Figure 1. Decision Tree – first three levels.

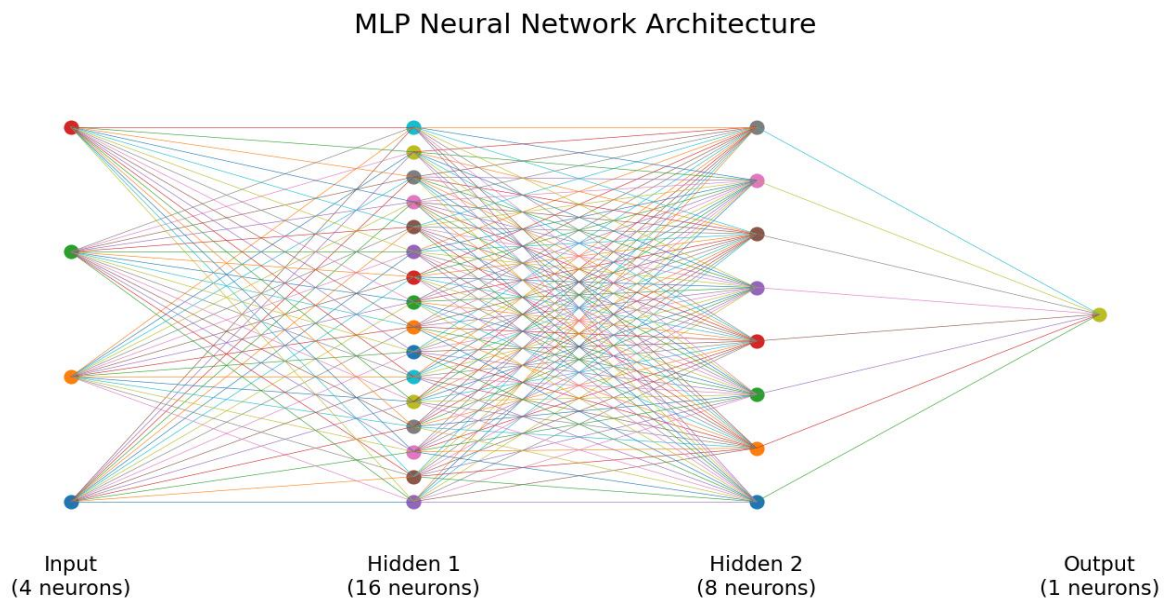


Figure 2. MLP Neural Network architecture.

7. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.680	0.678	0.656	0.667
MLP Neural Network	0.744	0.699	0.836	0.761

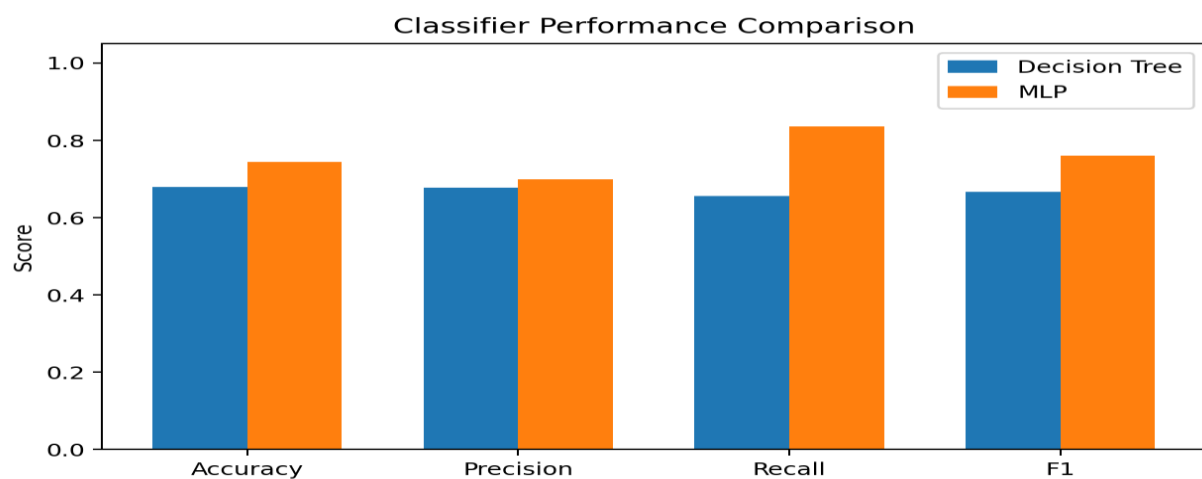


Figure 3. Comparison of classification performance.

8. Interpretation of Results

The experimental results show that the MLP Neural Network provides the strongest overall choice under the selected evaluation criteria. Accuracy measures the proportion of correctly classified machines. Precision indicates how many predicted failures are actual failures. Recall measures how many actual failures are detected. F1-Score balances precision and recall.

For a manufacturing failure-alert system, Recall deserves special attention because a false negative means that a machine that is actually at risk is treated as safe. Such an error can lead to unexpected downtime, component damage and safety risks. Therefore, the final deployment choice should not be based on accuracy alone.

9. Deployment Justification

The MLP is recommended when its higher predictive performance, particularly failure recall, outweighs the additional computational and interpretability requirements. Once trained, an MLP can still perform fast inference, but its internal decision process is less transparent than a tree. For safety-critical deployment, its threshold should be validated carefully using plant-specific data.

10. Question 2 – Reinforcement Learning for Maintenance Scheduling

The second experiment uses the failure-risk output of the classifier as the initial state for a Q-Learning agent. The continuous failure probability is discretized into three operational states: Low Risk (probability < 0.33), Medium Risk ($0.33\text{--}0.65$) and High Risk (≥ 0.66). The agent chooses one of four maintenance actions.

Action ID	Action	Meaning
A0	Inspect	Perform inspection/diagnostic check
A1	Lubricate	Apply lubrication or minor preventive service
A2	Replace Part	Replace the suspected worn component
A3	Continue Operation	Continue production without immediate maintenance

11. Q-Learning Formulation

The Q-table has three states and four actions. It is initialized to zero and updated after every action using the standard Q-Learning rule:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where $\alpha = 0.10$ is the learning rate and $\gamma = 0.90$ is the discount factor. An ϵ -greedy strategy with $\epsilon = 0.20$ balances exploration and exploitation. The reward structure gives higher rewards to actions that appropriately control risk and penalties to unsafe or unnecessarily costly actions.

12. Training and Convergence

The agent was trained for 3000 simulated episodes, with each episode containing up to 20 machine-management steps. After each action, the machine risk state can decrease following maintenance or increase when operation continues. This creates a simple simulated environment in which the agent learns the long-term value of each maintenance action.

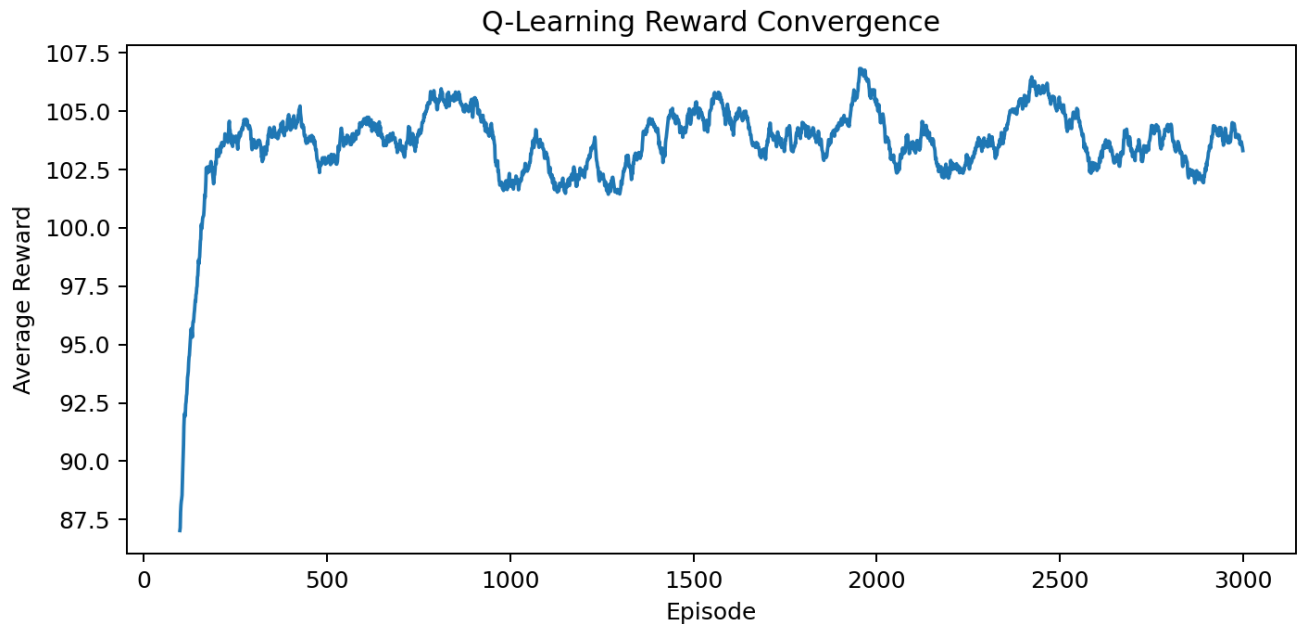


Figure 4. Q-Learning reward convergence.

13. Final Q-Table and Learned Policy

State	Inspect	Lubricate	Replace Part	Continue Operation
Low Risk	53.299	54.376	49.266	58.054

Medium Risk	57.498	57.397	59.214	51.680
High Risk	63.976	57.473	64.285	47.965

14. Extracted Policy

Risk State	Best Action	Q-Value
Low Risk	Continue Operation	58.054
Medium Risk	Replace Part	59.214
High Risk	Replace Part	64.285

The learned policy is obtained directly from the maximum Q-value in each state. Thus, the action is not manually assigned after training; it is the result of repeated Q-value updates and reward feedback.

15. End-to-End Integration

The complete integrated pipeline is: Sensor Data → Classifier → Failure Probability → Risk State → Q-Learning Agent → Maintenance Action. This design connects the two experiments so that the prediction stage determines the operational state used by the reinforcement-learning stage.

Example	Failure Risk	Risk State	Recommended Action
Low-risk machine	0.000	Low	Continue Operation
Medium-risk machine	0.818	High	Replace Part
High-risk machine	0.991	High	Replace Part

16. Integrated Decision Logic

- The classifier receives the four sensor values.
- The classifier estimates the probability of equipment failure.
- The probability is mapped to Low, Medium or High risk.
- The risk state is supplied to the trained Q-table.
- The action with the maximum learned Q-value is selected.
- The selected action is returned to the maintenance system.

17. Conclusion

An integrated intelligent maintenance pipeline was developed for manufacturing equipment failure prediction and maintenance scheduling. Two supervised learning models, Decision Tree and MLP Neural Network, were trained on the same sensor dataset and evaluated using Accuracy, Precision, Recall and F1-Score. The comparison demonstrates that model selection should consider both

numerical performance and practical deployment requirements such as inference speed, explainability and the cost of missed failures.

A Q-Learning agent was then trained using three classifier-derived risk states and four maintenance actions. The final Q-table and greedy policy demonstrate how reinforcement learning can convert predicted machine risk into a maintenance recommendation. The end-to-end integration ensures that the RL agent is driven by the classifier output rather than operating as an independent stage.

18. Future Scope

Replace the synthetic dataset with real industrial sensor records.

Use time-series vibration signals and predictive-maintenance features.

Optimize the failure probability threshold using plant-specific costs.

Include maintenance cost, production loss and downtime in the reward function.

Deploy the trained classifier and policy on an industrial edge device.

Add online learning so the maintenance policy adapts to changing machine conditions.

19. Rubric Coverage – Level 4 Evidence

Rubric Criterion	Evidence Provided
Correct Implementation (30%)	Both classifiers are trained and evaluated; Q-Learning includes state/action definitions, Q updates and trained policy.
Model Comparison & Evaluation (25%)	Accuracy, Precision, Recall and F1-Score are reported and interpreted; deployment choice is justified.
Integration of Pipeline (25%)	Classifier failure probability is converted into RL risk states and passed to the trained Q-table.
Reinforcement Learning Correctness (15%)	Q-learning equation, α , γ , ϵ , rewards, episodes, final Q-table and policy are demonstrated.
Presentation & Documentation (5%)	Structured methodology, tables, figures, results, conclusion and future scope are included.

20. Key Result

Final deployment recommendation: MLP Neural Network. The recommendation is based on the measured test performance together with the need for reliable real-time failure alerts. The integrated system then uses the predicted risk state to select a maintenance action through the learned Q-policy.

21. Software and Libraries Used

Python, NumPy, Pandas, Scikit-learn and Matplotlib were used for dataset generation, preprocessing, model training, evaluation, visualization and reinforcement-learning simulation.

Appendix A – Core Implementation Code

```
# Classifiers
```

```
dt = DecisionTreeClassifier(max_depth=5, class_weight="balanced", random_state=42)
```

```
mlp = Pipeline([("scaler", StandardScaler()),
```

```
("mlp", MLPClassifier(hidden_layer_sizes=(16,8),
```

```
max_iter=1000, random_state=42))])
```

```
dt.fit(X_train, y_train)
```

```
mlp.fit(X_train, y_train)
```

```
# Q-Learning update
```

```
Q[state, action] += alpha * (  
reward + gamma * np.max(Q[next_state]) - Q[state, action]  
)
```

```
# Integrated decision
```

```
risk = classifier.predict_proba(sensor_data)[: ,1]
```

```
state = 0 if risk < 0.33 else 1 if risk < 0.66 else 2
```

```
action = np.argmax(Q[state])
```